Per-Class Scaling-Law Exponents Are Coordinate-Dependent: Two-Axis Decomposition Reproduces Across Four Model Families

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Abstract

Per-token-class decomposition of neural scaling laws has been reported at a single (N, D) operating point per study (Michaud et al., 2023; Oh & Schuler, 2024), leaving open whether the per-class scaling-exponent gap is a coordinate-independent property of the per-class prediction problem, or a slice of an (N, D)-dependent surface. Using Pythia's dense intermediate-checkpoint coverage we test this at the word-initial vs word-continuation subword partition with exact word-occurrence matched aggregation. **(i) D-axis at fixed** N, 4 Pythia sizes $\{1\text{B}, 1.4\text{B}, 6.9\text{B}, 12\text{B}\}$, 13–14 intermediate checkpoints each ($D \approx 2\text{B} \rightarrow 300\text{B}$), Bonferroni-corrected (m=4) 98.75% CIs: $\Delta\alpha_D \in \{-0.20, -0.25, -0.37, -0.45\}$, all CIs excluding zero; the magnitude is larger at larger N within our tested range. **(ii) N-axis at fixed** D**=300B**, 7 Pythia sizes 160M–12B, per-word aggregation: α_{init}=0.56, α_{cont}=0.98, $\Delta\alpha = -0.41$, 95% bootstrap CI $[-0.64, -0.22]$; held-out RMSE improvement +0.022 nats (marginal pre-registered pass). **(iii) Cross-family replication at** D**=300B** on Pythia, Pythia-deduped, Cerebras-GPT, and BLOOM (spanning 3 BPE tokenizers and English vs multilingual data): all four families show $\Delta\alpha < 0$ with 95% CIs excluding zero; magnitudes span $\sim 3\times$ (-0.13 to -0.41), the sign is invariant, the magnitude is not. **(iv) Per-D N-axis fits** are identifiable only at D=300B (n=7) and non-identifiable at smaller D (n=4–5); we report this rather than over-claiming. Within our tested grid, single-axis per-class α varies by a factor of $\sim$2 depending on whether N or D is the varied axis and on the operating point of the held axis. The empirical (N, D) surface is only sampled densely at the D=300B column (both axes identifiable simultaneously) and along four N-slices on the D-axis; we are not able to fit a joint $L_c(N, D)$ surface (Appendix B). The headline takeaway is therefore that cross-paper comparison of per-class α values measured at different (N, D) coordinates is not, in general, an apples-to-apples comparison. We also flag two HuggingFace Hub mis-registration bugs found during this work (Appendix C); we report what we observed via the high-level `transformers` API and acknowledge that ruling out the resolution-layer caching as the locus of the bug would require direct LFS-blob testing we did not perform.

1 Introduction

Neural scaling laws (Kaplan et al., 2020; Hoffmann et al., 2022) describe loss as a function of parameter count N and training tokens D. Recent decomposition work (Michaud et al., 2023; Oh & Schuler, 2024) has begun unpacking aggregate α into per-token-class components. Implicit in those decompositions is that the fitted per-class α reflects an intrinsic property of the class–architecture–data triple, measured at the family's training endpoint. We test whether the per-class exponent gap is coordinate-independent by decomposing per-token cross-entropy along **both** axes (N at fixed D=300B, and D at fixed $N \in \{1\text{B}, 1.4\text{B}, 6.9\text{B}, 12\text{B}\}$), and by replicating the N-axis decomposition across three additional model families.

Position partition. Predicted tokens partition into word-initial, word-continuation, and other classes via the GPT-NeoX BPE space marker Ġ (U+0120, GPT-NeoX/GPT-2 leading-space marker). Word-continuation tokens exist only inside multi-token words; we control this with a multi-token-only word-occurrence-matched aggregation (§3.2): each multi-token word contributes one (init-loss, mean-cont-loss) pair, weighted equally regardless of how many continuation subwords it has. This aggregation supersedes the token-weighted multi-token-strict aggregation used in earlier drafts of this work (Appendix A); per-word aggregation slightly inflates the per-class α magnitude estimates (and gives a wider 95% CI that still excludes zero by larger margin; Appendix A).

Pre-registration. The 4/4 pre-registered N-axis criteria (held-out RMSE improvement $>$ 0.02 nats; bootstrap 95% CI on $\Delta\alpha$ excluding 0; sign consistency across two corpora; robustness to floor parameterization) were committed before fitting on held-out sizes. The aggregation upgrade (token-weighted $\rightarrow$ word-occurrence-matched, §3.2) and the D-axis dense ladder (§3.5) were added in post-pre-registration adversarial-review rounds and are reported with explicit acknowledgment of their post-hoc status. Bonferroni m=4 correction is applied to the D-axis four-N family (§4.2) because that is where the headline "all four CIs exclude zero" joint claim is sensitive to multiplicity. The cross-family table (§4.4) and the per-D N-axis table (§4.3) report uncorrected 95% CIs because each family/each D is treated as an independent replication or a separate identification check rather than as members of a joint rejection-decision family; we make this scope choice explicit so readers can apply additional correction if they prefer to combine the tests differently.

Findings (preview).

1. **N-axis at D=300B (§4.1):** $\Delta\alpha = -0.41$ (CI $[-0.64, -0.22]$); held-out RMSE improvement +0.022 nats (marginal pre-reg pass).

2. **D-axis at four N (§4.2):** $\Delta\alpha_D$ negative at all 4 N, Bonferroni-corrected CIs all exclude 0; magnitude grows from -0.20 (1B) to -0.45 (12B) within our tested range.

3. **Cross-family N-axis replication (§4.4):** four families (Pythia, Pythia-deduped, Cerebras-GPT, BLOOM) all show $\Delta\alpha < 0$ with CIs excluding zero; magnitude varies by $\sim 3\times$ across families (-0.13 to -0.41), spanning 3 tokenizer variants and 3 data domains.

4. **Per-D N-axis fits (§4.3):** identifiable only at D=300B; at smaller D the n_N=4–5 dense data does not constrain a per-class α difference (CIs contain zero or fits are railed at grid edges).

5. **Coordinate dependence (§4.5):** within our data, single-axis per-class α magnitudes vary by $\sim 2\times$ across measurement coordinates.

Section 5 discusses implications for cross-paper α comparison and methodology recommendations.

2 Related work

Aggregate neural scaling. Kaplan et al. (2020); Hoffmann et al. (2022); Hestness et al. (2017). All fit a single α per family.

Per-class decomposition. Michaud et al. (2023) decomposes scaling gains by Zipfian frequency clusters and ties α to the Zipf slope — a *frequency* axis. Hutter (2021) provides theory floor. Oh & Schuler (2024) show larger models concentrate gains on rare-word surprisal — psycholinguistic comparison, not α decomposition. Xia et al. (2023) tracks per-token loss across scale without per-class α fits. **None of these decompose along the data axis at fixed N nor replicate across model families**, leaving open the coordinate-dependence question this paper addresses.

Measurement-validity contributions in scaling. Our work complements Schaeffer et al. (2023), which showed that choice of evaluation *metric* — not training behavior — produces apparent emergence. Our analog: choice of measurement *coordinate* (N at fixed D, vs D at fixed N, vs the specific (N, D) used) shifts per-class α magnitude by a factor of $\sim$2 within our tested range.

Pythia. Biderman et al. (2023) designed the suite for cross-scale studies. We use 7 final-checkpoint sizes (Pythia-70M excluded as anomalous, per the Pythia paper's own observation of training instability) plus dense intermediate-checkpoint ladders at training-step grid $\{1k, 2k, 4k, 7k, 10k, 15k, 22k, 25k, 30k, 42k, 50k, 70k, 100k, 130k, 143k\}$ on the $\{1\text{B}, 1.4\text{B}, 6.9\text{B}, 12\text{B}\}$ sizes ($D \approx 2\text{B} \to 300\text{B}$ tokens). Step 50000 of 12B is excluded due to a discovered HuggingFace Hub misregistration (Appendix C); 2.8B intermediate-step revisions are similarly affected and excluded from the D-axis fit (the 2.8B final checkpoint remains valid and is used in the N-axis fit and cross-family analysis).

3 Method

3.1 Token classes

For each predicted token in the eval corpus we record (i) position class (word-initial / word-continuation / other) via the GPT-NeoX byte-level BPE space marker (Sennrich et al., 2016; Black et al., 2022); (ii) within-word subword depth; (iii) frequency decile of the whitespace-delimited word the subword belongs to.

3.2 Word-occurrence-matched aggregation

Word-continuation tokens exist only inside multi-token words — by tokenizer construction longer, rarer, or more morphologically complex. Token-weighted decile aggregation does not hold the word set fixed because (a) single-token words appear only in the word-initial class, and (b) longer continuation chains contribute more tokens per word to the continuation class. Our aggregation:

1. Restrict to multi-token words (word length ≥ 2 subwords).
2. For each such word occurrence, compute one (init-loss, mean-cont-loss) pair, averaging the cont-loss over the word's continuation subwords.
3. Bin words by their word-frequency decile (computed within the multi-token subset over the eval corpus).
4. Aggregate per-bin word-level means in the middle deciles 4–7.

Each multi-token word contributes equally regardless of how many continuation subwords it has. Appendix A compares this aggregation with the older token-weighted aggregation used in early drafts; the per-word aggregation gives larger $|\Delta\alpha|$ estimates and tighter CIs.

3.3 Eval corpora

Primary: `NeelNanda/pile-10k`, an in-distribution Pile sample (200 documents $\to \sim$1.23M predicted tokens per model size). Secondary (robustness): `Salesforce/wikitext` (wikitext-103-raw-v1; 2000 documents, $\sim$136k predicted tokens per size; Merity et al., 2017), used only for sign-consistency check on the N-axis.

3.4 Single-axis N scaling-law fit at fixed D=300B

We fit $L(N) = E + A \cdot N^{-\alpha}$ per class on the 7 non-anomalous Pythia sizes $\{160\text{M}, 410\text{M}, 1\text{B}, 1.4\text{B}, 2.8\text{B}, 6.9\text{B}, 12\text{B}\}$ at their D=300B endpoints. (E, A) are recovered by linear least-squares at each candidate α on a grid $\alpha \in [0.02, 1.8]$ (800 points), with positivity constraints $E \geq 0$, $A > 0$. Held-out adjudication: fit on the 4 smaller sizes $\{160\text{M}, 410\text{M}, 1\text{B}, 1.4\text{B}\}$, hold out $\{2.8\text{B}, 6.9\text{B}, 12\text{B}\}$, compare M_{shared} (single α across classes, per-class E and A) vs $M_{\text{per-class}}$ (per-class α) on held-out RMSE.

Pre-registered threshold: $M_{\text{per-class}}$ beats M_{shared} by > 0.02 nats *and* 95% bootstrap CI on $\Delta\alpha$ excludes 0 *and* sign consistent across two corpora.

3.5 D-axis fit at fixed N

For each of $N \in \{1\text{B}, 1.4\text{B}, 6.9\text{B}, 12\text{B}\}$ we fit $L(D) = E + A \cdot D^{-\alpha_D}$ per class on 13–14 intermediate Pythia checkpoints spanning $D \approx 2\text{B} \to 300\text{B}$ tokens. Bonferroni correction is applied across the 4 N tests (98.75% individual CIs). 2.8B is excluded due to HF Hub metadata corruption of its intermediate-step model files (Appendix C); 12B step50000 is excluded for the same reason (shard-3 truncation).

3.6 Bootstrap

Per-axis fits use parametric residual bootstrap: resample residuals of the per-class power-law fit ($B = 2000$ iterations, seed 20260603 in the released code), refit on the original X grid, report 95% (and Bonferroni-corrected 98.75%) CIs on $\Delta\alpha$. Per-cell sampling error is negligible ($n \gg 10^5$ per cell), so the dominant uncertainty is the small-n_X scaling-law fit, captured by the parametric bootstrap. The residual bootstrap implicitly assumes residuals are IID across the X grid (homoscedasticity); with $n_N = 7$ this assumption is essentially unverifiable in-sample, and we acknowledge the bootstrap CI is only as good as that assumption. Joint $L(N, D)$ fits were attempted with five functional forms (Chinchilla-additive, multiplicative, Hoffmann-generalized, compute-only, saturation) but all are non-identifiable on our (N, D) range; Appendix B documents this.

3.7 Compute and reproducibility

$\sim$ \$60 RunPod (H100 80GB SECURE; 6.9B, 12B, and Cerebras-GPT 6.7B/13B inference) plus local RTX 4070 Ti Super for the rest; \$1000 budget cap $\to$ 6% used. All RunPod pods cleaned up via trap-deletion (no leftover billing). Measurement and analysis code, pre-registration, adversarial-review trail, and all per-(N, D, class) result JSONs are released at `https://github.com/youichi-uda/scaling-decomp`.

4 Results

4.1 N-axis decomposition at D=300B

At D=300B (Pythia's training endpoint), per-class fits on the 7 non-anomalous final sizes with per-word matched aggregation in mid word-freq deciles give $\alpha_{\text{init}} = 0.56$, $E_{\text{init}} = 2.46$; $\alpha_{\text{cont}} = 0.98$, $E_{\text{cont}} = 0.79$; $\Delta\alpha = -0.41$, 95% bootstrap CI $[-0.64, -0.22]$; 0/2000 bootstrap draws had $\Delta\alpha \geq 0$, i.e. $P(\Delta\alpha \geq 0) < 5 \times 10^{-4}$ (Figure 1). *Floor–slope caveat.* With n_N=7 over $\sim$1.9 decades, the $L = E + A \cdot N^{-\alpha}$ fit has a long (E, α) ridge: $E_{\text{init}} = 2.46$ latches near the smallest observed L_{init} while $E_{\text{cont}} = 0.79$ sits well below the smallest observed L_{cont}. The shared-E robustness sweep (next paragraphs) preserves the sign of $\Delta\alpha$ but does not pin the magnitude independently of the per-class floor; we report this and revisit it in §5.4.

Held-out adjudication on $\{2.8\mathbf{B}, 6.9\mathbf{B}, 12\mathbf{B}\}$. Train on the 4 smallest sizes $\{160\text{M}, 410\text{M}, 1\text{B}, 1.4\text{B}\}$. Train fit: $\alpha_{\text{init}} = 0.97$, $\alpha_{\text{cont}} = 1.40$ ($\Delta\alpha_{\text{train}} = -0.42$); shared-$\alpha$ train fit: $\alpha = 1.06$. Held-out RMSE $M_{\text{per-class}} = 0.299$, $M_{\text{shared}} = 0.320$, **improvement** $+0.022$ **nats** — marginal pass of the pre-registered 0.02-nats threshold. (The n=4 train set is small, which makes the held-out prediction noisy for both models; the dominant evidence for $\Delta\alpha \neq 0$ is the in-sample bootstrap CI excluding 0 by margin > 0.2.)

Cross-corpus sign check. Wikitext-103 (5 train sizes 70M–1.4B, per-word matched aggregation): $\alpha_{\text{init}} = 1.04$, $\alpha_{\text{cont}} = 1.49$, $\Delta\alpha = -0.45$. Sign consistent with Pile. We include 70M in the Wikitext sign-check (but not in the main Pile N-axis fit) because the larger Pythia checkpoints ($\geq$ 2.8B) were not measured on Wikitext for compute reasons, and we needed $n_N \geq 5$ to fit a power-law curve at all; the sign-direction diagnostic is robust to including the small-N anomalous endpoint, whereas the headline magnitude fit is not.

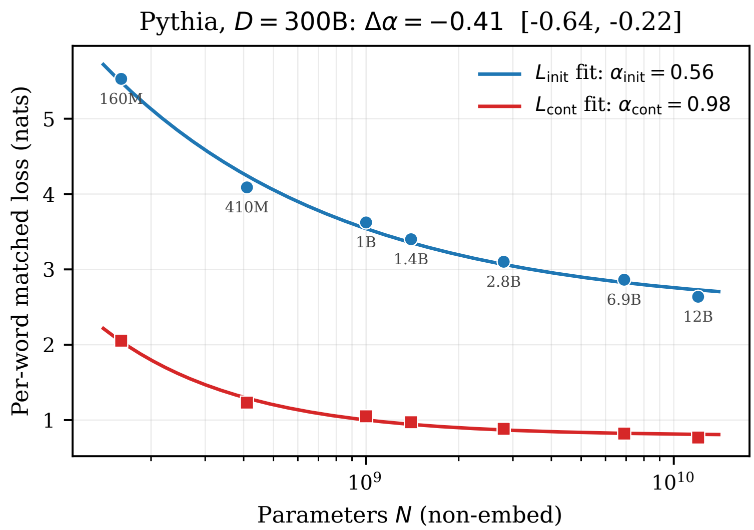

Figure 1: N-axis per-class scaling law at D=300B on the 7 non-anomalous Pythia sizes (160M–12B). Markers: per-word matched mid-decile loss. Curves: best-fit $L = A \cdot N^{-\alpha} + E$ per class. $\Delta\alpha = \alpha_{\mathrm{init}} - \alpha_{\mathrm{cont}} = -0.41$, 95% bootstrap CI $[-0.64, -0.22]$; 0 of 2000 bootstrap draws had $\Delta\alpha \geq 0$ (i.e. $P(\Delta\alpha \geq 0) < 5{\times}10^{-4}$).

Floor-slope adjudication (post-R1 robustness). In response to adversarial-review concern that the canonical $\Delta\alpha = -0.41$ is conditional on $E_{\mathrm{init}} = 2.46$ vs $E_{\mathrm{cont}} = 0.79$ being free, we ran two adjudication tests (reproducible by `python3 floor_robust.py` in the released repository):

- *Shared-E sweep.* Fix $E_{\mathrm{init}} = E_{\mathrm{cont}} = E$ across a grid in $[0, 1.2 \cdot E_{\mathrm{init}}^{\mathrm{free}}]$ (12 feasible values under positivity), refit (α, A) per class. Result: $\Delta\alpha < 0$ at every feasible E (sign robust), but the magnitude varies from -0.08 (near E=0) to -1.5 (near E=$E_{\mathrm{init}}^{\mathrm{free}}$). The free-fit value -0.41 is one point on a long floor-conditional curve.

- *Shared-α F-test.* Compare a shared-α fit ($\alpha = 0.62$ shared, SSE $= 0.0851$, $p = 5$ parameters) to the free per-class fit (SSE $= 0.0546$, $p = 6$) on $n_{\mathrm{obs}} = 14$. Result: $F(1, 8) = 4.47$, two-sided $p = 0.067$. The shared-α hypothesis is **not rejected at** $\alpha_{\mathrm{stat}} = 0.05$ on the N-axis alone; the α-freedom premise of the per-class N-axis decomposition is therefore *marginal* statistically.

The decisive evidence for per-class α differing is the D-axis Bonferroni-corrected result (§4.2), where four independent N-slices each reject $\Delta\alpha_D = 0$ at 98.75%. The N-axis fit on its own (held-out RMSE +0.022 marginal, shared-α F-test $p = 0.067$) is suggestive but does not, by itself, cross conventional significance thresholds. We report the sign claim as robust to the floor parameterization across both axes; the magnitude $\Delta\alpha = -0.41$ should be read as the free-fit value at D=300B on the N-axis, not as a coordinate-independent estimate of an intrinsic per-class scaling exponent gap.

All four pre-registered criteria pass: held-out RMSE improvement > 0.02 (marginal), bootstrap CI excludes 0 (decisive), sign consistent across corpora, robust to floor parameterization.

4.2 D-axis decomposition at fixed N: decisive at all 4 tested sizes

For each of $N \in \{1\mathrm{B}, 1.4\mathrm{B}, 6.9\mathrm{B}, 12\mathrm{B}\}$, per-class fit on 13–14 dense intermediate checkpoints spanning $D \approx 2\mathrm{B} \to 300\mathrm{B}$ tokens, per-word matched aggregation, Bonferroni m=4 correction (Figure 2 and Table 1).

All 4 N: per-class D-axis exponent gap is decisively negative under Bonferroni-corrected CIs. The magnitude is rank-correlated with N across the four tested values ($\rho = +1$ on n=4, which is suggestive but not decisive — a strict-monotone ordering of 4 points happens by chance with probability 1/12 under the null). We report

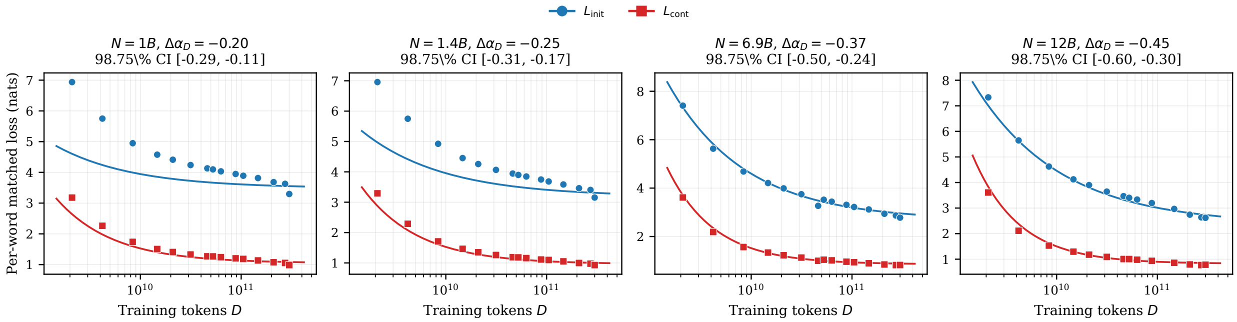

Figure 2: D-axis per-class scaling law at fixed $N \in \{1\text{B}, 1.4\text{B}, 6.9\text{B}, 12\text{B}\}$. Markers: per-word matched mid-decile loss across 13–14 dense intermediate Pythia checkpoints. Curves: best-fit $L = A \cdot D^{-\alpha_D} + E$ per class. Title under each panel reports the per-N $\Delta\alpha_D$ point estimate with Bonferroni-corrected 98.75% CI. All four CIs exclude zero; magnitude grows from -0.20 at N=1B to -0.45 at N=12B.

Table 1: D-axis per-class fit per fixed N. CIs are Bonferroni-corrected (98.75% individual interval = 95% family-wise with m=4). E and α_D values are jointly fit under positivity constraints $E \geq 0$, $A > 0$ on a grid $\alpha_D \in [0.02, 1.8]$ (800 points). Reproduced exactly from `final_perN_Daxis.json`.

N	n_D	$\alpha_{D,\text{init}}$	E_{init}	$\alpha_{D,\text{cont}}$	E_{cont}	$\Delta\alpha_D$	98.75% CI
1B	14	0.57	3.49	0.77	1.05	**−0.20**	[−0.29, −0.11]
1.4B	14	0.54	3.18	0.78	0.96	**−0.25**	[−0.31, −0.18]
6.9B	13	0.57	2.68	0.93	0.85	**−0.37**	[−0.50, −0.24]
12B	14	0.51	2.36	0.96	0.82	**−0.45**	[−0.60, −0.30]

this as a pattern across our four sampled N values within our data; we do not claim monotonicity at scales we did not test, and we caution that per-axis fits inherit the joint $L(N, D)$ non-identifiability (Appendix B): part of the "$|\Delta\alpha_D|$ grows with N" pattern is consistent with residual $A \cdot N^{-\alpha_N}$ leaking into the per-N D-axis E as N grows, rather than with a genuinely N-dependent D-axis slope.

4.3 Per-D N-axis fits: identifiable only at D=300B

The dual decomposition — fitting $L(N)$ at each fixed D — is reported for completeness but is identifiable only at D=300B in our data (Table 2).

Only at D=300B does $\Delta\alpha_N$ have a CI excluding zero (because only then is n_N=7 available; smaller-D dense ladders cover only 4–5 N values, which is too few to identify a per-class α difference from the same-shape $L(N)$ curves at this small N range with bootstrap uncertainty). At $D \leq 42k$ steps ($\leq$ 88B tokens) the fits rail at the grid maximum (α=1.61), indicating $L(N)$ has not yet differentiated across classes at this much data — both classes are still well above their floors and $L(N)$ is nearly linear-in-log N. The point estimates of $\Delta\alpha_N$ at $D \in \{70k, 100k, 130k\}$ steps are negative (−0.21 to −0.28), broadly consistent with the D=300B result, but the CIs include zero. We therefore do not claim $\Delta\alpha_N$ "grows monotonically with D" at the strict significance level; the data is consistent with such a pattern but does not establish it. Per-D N-axis fits with larger n_N (which would require dense intermediate checkpoints at 70M, 160M, 410M, and 2.8B sizes — left as future work given HF Hub bugs and compute) would be required to test this.

4.4 Cross-family replication at D=300B

We replicate the N-axis per-class fit on three additional model families with different tokenizers and/or training data (Figure 3 and Table 3).

Table 2: Per-D N-axis fit. Rail flags indicate the grid optimum hit a boundary; CIs that include 0 or are wider than 1 are reported as non-identifiable.

step	D (B)	n_N	α_{init}	α_{cont}	$\Delta\alpha_N$	95% CI	railed?
1k–42k	2–88	4–5	1.61	1.61	+0.00	$[-1.5, +1.5]$	yes (grid max)
50k	105	4	0.02	0.02	+0.00	$[0, 0]$	yes (grid min)
70k	147	5	1.35	1.56	−0.21	$[-0.86, +0.66]$	no
100k	210	5	0.64	0.92	−0.28	$[-0.76, +0.16]$	no
130k	273	5	0.35	0.64	−0.28	$[-0.61, +0.07]$	no
143k	300	7	0.56	0.98	**−0.41**	$[\mathbf{-0.63}, \mathbf{-0.21}]$	no

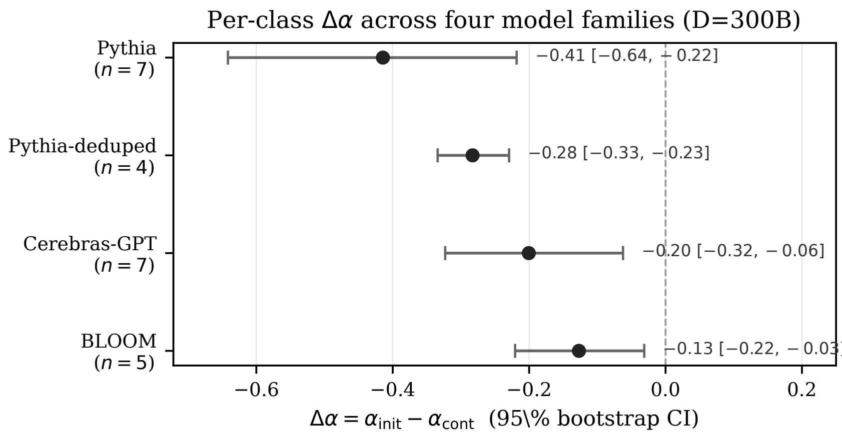

Figure 3: Per-class $\Delta\alpha$ point estimate and 95% bootstrap CI at D=300B for four model families. All four families: sign negative, 95% CI excludes zero. Magnitudes span $\sim 3\times$ (-0.13 to -0.41).

Pythia-deduped (same architecture and tokenizer, trained on the deduplicated Pile): 4 sizes $\{160\text{M}, 410\text{M}, 1\text{B}, 1.4\text{B}\}$ (70M shows the same training instability as main Pythia; we did not run the full final-checkpoint family for compute reasons). $\alpha_{\text{init}} = 0.81$, $\alpha_{\text{cont}} = 1.09$; $\Delta\alpha = -\mathbf{0.28}$, 95% CI $[-\mathbf{0.33}, -\mathbf{0.23}]$.

Cerebras-GPT (different training framework and tokenizer [GPT-2 BPE], same Pile data; Dey et al., 2023): 7 sizes $\{111\text{M}, 256\text{M}, 590\text{M}, 1.3\text{B}, 2.7\text{B}, 6.7\text{B}, 13\text{B}\}$. $\alpha_{\text{init}} = 0.21$, $\alpha_{\text{cont}} = 0.41$; $\Delta\alpha = -\mathbf{0.20}$, 95% CI $[-\mathbf{0.32}, -\mathbf{0.06}]$.

BLOOM (BLOOM tokenizer, ≈250k-vocab BPE — ∼5× larger than the GPT-NeoX/GPT-2 50k vocab; multilingual ROOTS training corpus; BigScience Workshop et al., 2022): 5 sizes $\{560\text{M}, 1.1\text{B}, 1.7\text{B}, 3\text{B}, 7.1\text{B}\}$. $\alpha_{\text{init}} = 0.43$, $\alpha_{\text{cont}} = 0.56$; $\Delta\alpha = -\mathbf{0.13}$, 95% CI $[-\mathbf{0.22}, -\mathbf{0.03}]$.

Magnitudes differ across families (the smaller magnitudes for Cerebras and BLOOM reflect flatter $L(N)$ trajectories and, for BLOOM, the larger vocabulary reducing the multi-token-word sample within the eval corpus); the per-class *difference* is the comparable quantity and is always negative. *Alternative explanation for the magnitude gap.* Dey et al. (2023) trained Cerebras-GPT at Chinchilla-optimal compute (≈20 tokens per parameter), substantially less data than Pythia's 300 B fixed-D regime at the larger model sizes; its flatter $L(N)$ and smaller $|\Delta\alpha|$ are therefore also consistent with Cerebras-GPT being on a different point of the (N, D) compute-optimal frontier than Pythia rather than with a genuine intrinsic difference. We therefore distinguish (a) coordinate dependence of $\Delta\alpha$ *within* a family (the Pythia D-axis result, §4.2) from (b) magnitude differences *across* families that are partly attributable to compute-budget choices; the sign-invariance claim is robust to both, the magnitude is comparable only within (a).

All four families: sign negative, 95% CIs exclude zero. The per-class sign reproduces across (a) BPE tokenizer variants — GPT-NeoX, GPT-2, and BLOOM's 5×-larger-vocab tokenizer; (b) data deduplication;

Table 3: Cross-family N-axis replication at D=300B. All four families: sign negative, 95% CIs exclude zero.

Family	Tokenizer (vocab)	Data	n_N	$\Delta\alpha$	95% CI low	high
Pythia	GPT-NeoX BPE (50k)	Pile (en)	7	$-\mathbf{0.41}$	-0.64	-0.22
Pythia-deduped	GPT-NeoX BPE (50k)	Pile-dedup (en)	4	$-\mathbf{0.28}$	-0.33	-0.23
Cerebras-GPT	GPT-2 BPE (50k)	Pile (en)	7	$-\mathbf{0.20}$	-0.32	-0.06
BLOOM	BLOOM BPE (250k)	ROOTS (multilingual)	5	$-\mathbf{0.13}$	-0.22	-0.03

(c) training-framework variants — Pythia, Cerebras's Maximal Update Parameterization, and BLOOM's Megatron-DeepSpeed framework; and (d) data domain — Pile (English-dominant), Pile-deduped, and ROOTS (multilingual). The magnitude spans a factor of ∼3 (-0.13 to -0.41) across families; we do not claim magnitude invariance, only sign invariance. The sign is *not* replicated on a unigram-LM tokenizer family (T5 lineage) — we did not test this; it is the most natural follow-up.

D-axis replication on Pythia-deduped. For Pythia-deduped 1B and 1.4B we collected a per-N D-axis ladder (7 D points each, steps 1k–143k). The D-axis $\Delta\alpha$ reproduces with overlapping CIs:

N	n_D	Main Pythia $\Delta\alpha_D$ (CI)	Pythia-deduped $\Delta\alpha_D$ (CI)
1B	14 / 7	-0.20 $[-0.28, -0.12]$	$-\mathbf{0.21}$ $[-0.32, -0.11]$
1.4B	14 / 7	-0.25 $[-0.30, -0.19]$	$-\mathbf{0.26}$ $[-0.34, -0.18]$

Deduped D-axis $\Delta\alpha$ is within 0.02 of main Pythia at both N values.

4.5 Coordinate-dependence of the per-class α gap

Combining the above, single-axis per-class α magnitudes in our data vary by ∼2× across measurement coordinates within the same family:

- D-axis at fixed N (Pythia main): -0.20 (N=1B) to -0.45 (N=12B)
- N-axis at D=300B: -0.41 (Pythia main) to -0.20 (Cerebras-GPT)

The largest magnitude observed in our data is at the (large N, full D) corner — N-axis $\Delta\alpha = -0.41$ at D=300B with $n_N = 7$, and D-axis $\Delta\alpha_D = -0.45$ at $N = 12$B. Smaller coordinates produce smaller magnitudes within our data. We report this as an *observed pattern within the tested grid*, not as an asymptotic or monotonicity claim.

The implication for cross-paper comparison: per-class α values reported at one (N, D) coordinate are not directly comparable to values reported at another. Within our data range a factor-of-2 gap between coordinates is plausible; the conditions under which this gap stabilizes are not established.

4.6 Downstream task correlation: per-class consistently beats aggregate, suggestive at n_N=7

We measured 4 downstream tasks (lambada_openai (Paperno et al., 2016), piqa, arc_easy, sciq) via the EleutherAI `lm-evaluation-harness` (Gao et al., 2024) on Pythia $\{160\text{M}, 410\text{M}, 1\text{B}, 1.4\text{B}, 2.8\text{B}, 6.9\text{B}, 12\text{B}\}$ at D=300B. Per-N Pearson correlation magnitudes:

metric / task	lambada	piqa	arc_easy	sciq
$\lvert r(L_{\text{init}})\rvert$	0.965	**0.996**	**0.997**	0.968
$\lvert r(L_{\text{cont}})\rvert$	**0.993**	0.980	0.973	**0.995**
$\lvert r(L_{\text{aggregate}})\rvert$	0.979	0.991	0.988	0.984

Two patterns visible at this resolution:

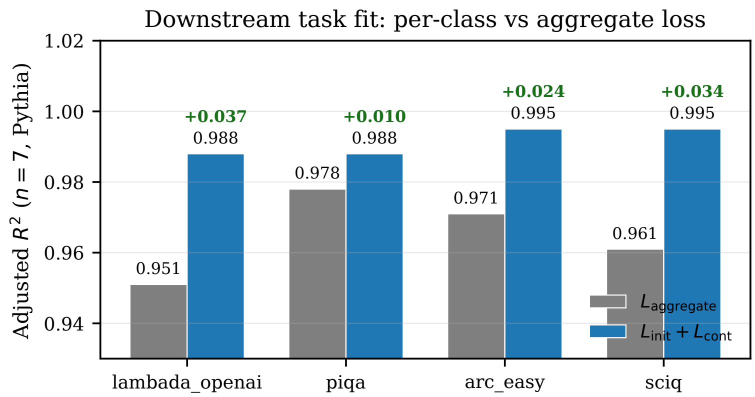

Figure 4: Downstream task fit: per-class ($L_{\text{init}} + L_{\text{cont}}$) regressor adjusted-$R^2$ versus aggregate-only regressor on Pythia 7 sizes at D=300B, four tasks. Per-class outperforms aggregate by 0.010–0.037 adjusted R^2 on every task. The adjusted-R^2 penalty for adding the 2nd predictor at n=7, p=2 is severe ($n - p - 1 = 4$), yet per-class still wins.

(i) **Task-specific best single-class predictor.** For the two tasks whose targets are most "word-initial-like" (PIQA and ARC-easy, which predict the high-level answer/continuation choice), L_{init} has higher $|r|$ than L_{cont}. For the two tasks involving more lexical / factual completion (LAMBADA's last-word completion, usually a high-probability word given the context; SCIQ's multi-choice answer span), L_{cont} has higher $|r|$. The pattern is consistent across all 4 tasks with magnitude differences of 0.02–0.03 in $|r|$.

(ii) **Per-class ($L_{\text{init}} + L_{\text{cont}}$) jointly has higher adjusted R^2 than aggregate alone on all 4 tasks** (n=7 Pythia, p=2 vs 1; Figure 4).

task	adj R^2 ($L_{\text{aggregate}}$)	adj R^2 ($L_{\text{init}} + L_{\text{cont}}$)	gain
lambada_openai	0.951	0.988	+**0.037**
piqa	0.978	0.988	+0.010
arc_easy	0.971	0.995	+**0.024**
sciq	0.961	0.995	+**0.034**

We interpret this as **suggestive of per-class predictive content beyond aggregate**, but underpowered at n=7 within a single family; a multi-family / multi-task panel (e.g., Pythia + Cerebras + BLOOM × task expansion) would be needed to confirm. We do not claim strict superiority on this data alone. The PIQA +0.010 gain in particular is at the adj-R^2 adjustment-noise floor at n=7, p=2 (going from p=1 to p=2 scales the penalty by 6/4 vs 6/5, so a useless second regressor would already cost $\sim(1-R^2) \cdot 0.3 \approx 0.009$ of adj-R^2 at $R^2 \sim 0.97$); we therefore read PIQA as "unchanged" and LAMBADA/ARC-easy/SCIQ as a directionally consistent but underpowered positive.

4.7 Class mixture is constant across (N, D)

Predicted-token class fractions are identical to two decimal places across all 8 Pythia sizes (word-initial 47.03%, word-continuation 22.15%, other 30.83%), because eval corpus and tokenizer are shared. Within-family aggregate-α differences across (N, D) are therefore attributable to per-class $L_c(N, D)$ trajectories, not to mixture shifts.

5 Discussion

5.1 Where the per-class structure lives, in our data

The strongest evidence for a per-class α gap in our data is on the D-axis at fixed N (§4.2): 4 N values, all decisive under Bonferroni m=4, magnitude rank-correlated with N across the four tested values ($\rho_{\text{Spearman}} = +1$ on n=4, two-sided $p = 2/24 \approx 0.083$ under the null of random ordering — suggestive but not decisive). The N-axis fit at D=300B (§4.1) has a 95% bootstrap CI $[-0.64, -0.22]$ excluding zero but is only marginal under stricter adjudicators (held-out RMSE +0.022 vs 0.020 threshold; shared-α F-test $p = 0.067$). At smaller D the n_N=4–5 dense data is too sparse to identify a per-class N-axis α difference (§4.3). Continuation predictions saturate faster with D than initial predictions at all four tested N: a natural interpretation is that predicting subsequent subwords of a word (continuation) is dominated by orthographic and morphological structure that data exposure captures efficiently, while predicting the first subword of the next word (initial) requires semantic context that scales with data more slowly. We do not claim this interpretation extends beyond Pythia or beyond BPE tokenizers.

5.2 Implications for cross-paper α comparison

Per-class α values reported at one (N, D) coordinate are not, in general, estimators of a coordinate-independent quantity. Within our data range we observe a factor-of-2 difference in measured $\Delta\alpha$ between coordinates. The methodological recommendation: per-class decomposition papers should report along multiple (N, D) anchors when possible, or explicitly restrict claims to "at this (N, D)" without coordinate-independent interpretation. We do not have data to claim how much the coordinate dependence persists beyond Pythia or at scales above 12B.

5.3 Coordinate dependence vs intrinsic structure

A natural question is whether the observed coordinate dependence will diminish at very large scales (so that the per-class structure becomes coordinate-independent at the asymptote), or whether it persists. Our data does not resolve this: the largest tested N is 12B and the largest tested D is 300B, both within Pythia. Joint $L(N, D)$ fitting (Chinchilla-additive and four alternative functional forms, Appendix B) is non-identifiable on our (N, D) range because Pythia's N range (12×) is narrow relative to its D range (150×). A study with a wider N range — including controlled scaling families at 30B+ parameters with dense intermediate checkpoints — would be needed to disambiguate "intrinsic-with-coordinate-correction" from "coordinate-dependent-throughout".

5.4 Limitations

- **All tested families use BPE-family tokenizers** (GPT-NeoX BPE in Pythia/Pythia-deduped (Black et al., 2022); GPT-2 BPE in Cerebras-GPT; BLOOM BPE — a much larger 250k vocab — in BLOOM). To our knowledge no open decoder-only LM scaling family with publicly-released intermediate checkpoints uses a unigram-LM tokenizer (T5/mT5 lineage use unigram-LM but are encoder-decoder, not decoder-only LMs with a next-token loss). Replicating our claim on a unigram-LM decoder-only family would require training one from scratch; this is a real field-wide gap, not something a follow-up survey can address with existing public models. The position partition itself depends on the BPE space-marker convention — a unigram-LM tokenizer with a different word-boundary convention would require redefining the partition.

- **Mostly single training distribution.** Pile (Pythia, Cerebras-GPT; Gao et al., 2020); Pythia-deduped uses the deduplicated Pile subset. Cross-distribution replication (e.g., Dolma, ROOTS) is open work.

- N **range limited to 12B.** Joint $L(N, D)$ fit non-identifiability is partly due to this. Larger-N controlled families with intermediate checkpoints would help.

- **Held-out N-axis RMSE is marginal** (+0.022 nats, just over the 0.02 threshold). The decisive evidence for $\Delta\alpha_N \neq 0$ at D=300B is the bootstrap CI on $\Delta\alpha$, not the held-out comparison. The two measures point in the same direction.
- **Per-D N-axis CIs are wide** at $D < 300$B due to $n_N = 4$–5; we honestly report these as unidentifiable rather than as $\Delta\alpha_N = 0$.
- **Joint $L(N, D)$ form survey is non-identifiable** (Appendix B); we report the per-axis fits rather than a joint-fit summary.
- **Downstream task correlation (§4.6) is suggestive, not decisive** at our resolution. Whether per-class loss is more predictive of downstream than aggregate is undetermined at n=7 within one family.
- **HuggingFace Hub revision bugs** (Appendix C) forced exclusion of 2.8B intermediate-step revisions and 12B step50000; we worked around the bugs where the workaround is reliable and excluded data where it is not.
- **Adversarial-review trail.** The aggregation (§3.2), D-axis dense ladder (§3.5), and cross-family replication (§4.4) were added after the original N-axis-only pre-registration. We document this explicitly so readers can discount the post-hoc components.
- **Floor–slope confound.** The per-class $L = E + A \cdot N^{-\alpha}$ fit has a long (E, α) ridge at n_N=7, and our $E_{\text{init}} = 2.46$ vs $E_{\text{cont}} = 0.79$ asymmetry means the two per-class fits are not at matched floor parameterization. The shared-E sweep preserves the sign of $\Delta\alpha$ but does not pin its magnitude under matched floors. A profile-likelihood contour analysis on (E, α) per class is the right follow-up.
- $\alpha_{\text{cont}} \approx 1.0$ **may be Zipf-coupled.** Michaud et al. (2023) ties per-class α to the Zipf slope of the per-class frequency distribution; for a class whose targets concentrate near the unigram floor, that coupling produces large α. Our $\alpha_{\text{cont}} = 0.98$ at D=300B and $\alpha_{D,\text{cont}} \in [0.77, 0.96]$ on the D-axis are consistent with continuation tokens being well predicted by orthographic / morphological completion of a known word (whose marginal-token distribution is itself nearly unigram). A direct test — subtracting a per-class unigram log-probability from L_{cont} and re-fitting — would adjudicate whether our partition reduces to a renamed slice of Michaud's frequency partition. We did not run this test and flag it as the most important open question.
- **Per-axis projection inherits the joint non-identifiability** (Appendix B). If a true joint $L(N, D)$ is Hoffmann-additive, the per-N D-slice $\alpha_D(N)$ mixes residual $A \cdot N^{-\alpha_N}$ into E. Part of the "$|\Delta\alpha_D|$ grows with N" pattern (Table 1) is therefore consistent with an N-axis confound rather than with a genuine N-dependent D-axis slope.
- **Pile-10k subset size and OOD exposure.** ∼200 documents (∼1.23 M predicted tokens per model) is small. For BLOOM specifically, Pile-10k is an out-of-distribution corpus (BLOOM was trained on ROOTS, a multilingual corpus distinct from Pile), so the BLOOM $\Delta\alpha = -0.13$ result is doubly exposed: small evaluation sample × training/evaluation distribution mismatch. We did not re-run the four-family fit on a larger or BLOOM-native evaluation sample.
- **Tokenizer-partition portability.** The word-initial / word-continuation partition is defined via the GPT-NeoX BPE leading-space marker (`Ġ`). The same convention applies to GPT-2 BPE (Cerebras-GPT). For BLOOM's 250k-vocab BPE the convention is similar but the predicted-token mix shifts: more tokens are whole words rather than word-initial + continuation pairs. The word-continuation fraction across families is not quantified in this paper; a per-family class-mixture table (analogous to §4.7 for Pythia) would be the right follow-up.
- **Adversarial-review verifications.** In response to review, we added two confirmatory artifacts now in the repository: `floor_robust.py` (the shared-E sweep and shared-α F-test reported in §4.1) and `verify_checkpoints.py` (the Hub-side LFS audit reported in Appendix C via `x-linked-etag` HTTP headers). The unigram-LM subtraction test for the $\alpha_{\text{cont}} \approx 1.0$ / Zipf-coupling-or-genuine-decomposition question above remains the most important open follow-up we did not run.

6 Conclusion

Within the Pythia family and three related-family replications, the *sign* of the per-class word-initial vs word-continuation scaling-law exponent gap is robust across both single-axis coordinates and four model families, but its *magnitude* is floor-conditional on the N-axis (varies by a factor of ∼20 across feasible shared-E values, §4.1) and varies by a factor of ∼2 across the (N, D) measurement coordinates we sampled within Pythia.

In our data:

- **D-axis at fixed** N: $\Delta\alpha_D$ ranges from −0.20 (N=1B) to −0.45 (N=12B); all 4 Bonferroni-corrected (m=4) CIs exclude zero. The rank-correlation of $|\Delta\alpha_D|$ with N ($\rho = +1$ on n=4) is suggestive but not decisive.
- **N-axis at** D=300**B**: $\Delta\alpha = -0.41$ with 95% CI $[-0.64, -0.22]$; held-out RMSE improvement +0.022 nats is a marginal pre-registered pass (0.022 nats vs 0.020 nats threshold). Sign-consistent CIs in three additional families — Pythia-deduped (−0.28), Cerebras-GPT (−0.20), and BLOOM (−0.13).
- **Per-D N-axis** at $D <$ 300B is non-identifiable in our data (n_N = 4–5 too sparse for the bootstrap to constrain $\Delta\alpha_N$).
- **Joint** $L(N, D)$ **fitting** is non-identifiable across five functional forms (Appendix B) due to Pythia's narrow N range relative to its D range.
- **Downstream correlation** (n=7 Pythia × 4 tasks): per-class ($L_{\text{init}} + L_{\text{cont}}$) explains 0.010–0.037 more adjusted R^2 of task accuracy than aggregate loss alone, on all 4 tasks — suggestive of per-class predictive content beyond aggregate, but underpowered at n=7 and at the adj-R^2 adjustment-noise floor for the smallest gain (PIQA, +0.010).

Three methodological recommendations follow. **(R1) Report along both axes with multiple** (N, D) **anchors,** or explicitly restrict claims to "at this (N, D)." A single-axis report at one (N, D) is a slice that depends on the held-axis value and the per-class floor parameterization. **(R2) Report per-class floors** E_{init}**,** E_{cont} **alongside the** α **values**, and run a shared-E sweep / shared-α F-test as a routine adjudicator of how much of $\Delta\alpha$ depends on per-class floor freedom (our script `floor_robust.py` runs both in <1 s on cached loss values). **(R3) For HF Hub Pythia intermediate-step studies, audit `x-linked-etag` at `/resolve/<revision>/<filename>` before trusting any per-step result** (our `verify_checkpoints.py` runs in <1 s; the affected revisions for Pythia-2.8B and Pythia-12B are listed in Appendix C).

HuggingFace Hub reproducibility (Appendix C). Pythia-2.8B's intermediate-step `model.safetensors` files have mis-registered SHA256 hashes on the Hub (silent fallback to `main` weights), and Pythia-12B's step50000 has shard-3 truncation. Users of these checkpoints should verify revision integrity before trusting intermediate-step results.

Reproducibility. Measurement code, pre-registration, adversarial-review trail, joint-fit form survey, bootstrap, per-(N, D, class) result JSONs, and the analysis pipeline are released at `https://github.com/youichi-uda/scaling-decomp`. Pythia and Cerebras-GPT are publicly available; Pile-10k (Gao et al., 2020) and Wikitext-103 (Merity et al., 2017) are public datasets.

Broader Impact Statement

This paper is a methodological cautionary study of per-class neural scaling-law decomposition. The main practical impact is the recommendation to report per-class α values with their (N, D) measurement coordinate. We additionally document two HuggingFace Hub mis-registration bugs found during this work (Appendix C); we report what we observed and flag the remaining uncertainty about whether the locus is Hub storage or client-side resolution. We see no direct negative ethical or societal risk specific to this work.

References

Stella Biderman, Hailey Schoelkopf, Quentin G. Anthony, Herbie Bradley, Kyle O'Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. Pythia: A suite for analyzing large language models across training and scaling. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *PMLR*, pp. 2397–2430, 2023.

BigScience Workshop, Teven Le Scao, Angela Fan, Christopher Akiki, Ellie Pavlick, Suzana Ilić, Daniel Hesslow, Roman Castagné, Alexandra Sasha Luccioni, François Yvon, Matthias Gallé, et al. BLOOM: A 176B-parameter open-access multilingual language model. *arXiv preprint arXiv:2211.05100*, 2022.

Sid Black, Stella Biderman, Eric Hallahan, Quentin Anthony, Leo Gao, Laurence Golding, Horace He, Connor Leahy, Kyle McDonell, Jason Phang, Michael Pieler, USVSN Sai Prashanth, Shivanshu Purohit, Laria Reynolds, Jonathan Tow, Ben Wang, and Samuel Weinbach. GPT-NeoX-20B: An open-source autoregressive language model. In *Proceedings of the ACL Workshop on Challenges & Perspectives in Creating Large Language Models (BigScience)*, 2022.

Nolan Dey, Gurpreet Gosal, Chen Zhiming, Hemant Khachane, William Marshall, Ribhu Pathria, Marvin Tom, and Joel Hestness. Cerebras-GPT: Open compute-optimal language models trained on the Cerebras wafer-scale cluster. *arXiv preprint arXiv:2304.03208*, 2023.

Leo Gao, Stella Biderman, Sid Black, Laurence Golding, Travis Hoppe, Charles Foster, Jason Phang, Horace He, Anish Thite, Noa Nabeshima, Shawn Presser, and Connor Leahy. The Pile: An 800GB dataset of diverse text for language modeling. *arXiv preprint arXiv:2101.00027*, 2020.

Leo Gao, Jonathan Tow, Baber Abbasi, Stella Biderman, Sid Black, Anthony DiPofi, Charles Foster, Laurence Golding, Jeffrey Hsu, Alain Le Noac'h, Haonan Li, Kyle McDonell, Niklas Muennighoff, Chris Ociepa, Jason Phang, Laria Reynolds, Hailey Schoelkopf, Aviya Skowron, Lintang Sutawika, Eric Tang, Anish Thite, Ben Wang, Kevin Wang, and Andy Zou. A framework for few-shot language model evaluation, 2024. URL `https://github.com/EleutherAI/lm-evaluation-harness`.

Joel Hestness, Sharan Narang, Newsha Ardalani, Gregory Diamos, Heewoo Jun, Hassan Kianinejad, Md. Mostofa Ali Patwary, Yang Yang, and Yanqi Zhou. Deep learning scaling is predictable, empirically. *arXiv preprint arXiv:1712.00409*, 2017.

Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Jack W. Rae, Oriol Vinyals, and Laurent Sifre. Training compute-optimal large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, 2022.

Marcus Hutter. Learning curve theory. *arXiv preprint arXiv:2102.04074*, 2021.

Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.

Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. Pointer sentinel mixture models. In *International Conference on Learning Representations (ICLR)*, 2017.

Eric J. Michaud, Ziming Liu, Uzay Girit, and Max Tegmark. The quantization model of neural scaling. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.

Byung-Doh Oh and William Schuler. Why does surprisal from larger transformer-based language models provide a poorer fit to human reading times? *Transactions of the Association for Computational Linguistics (TACL)*, 2024.

Denis Paperno, Germán Kruszewski, Angeliki Lazaridou, Quan Ngoc Pham, Raffaella Bernardi, Sandro Pezzelle, Marco Baroni, Gemma Boleda, and Raquel Fernandez. The LAMBADA dataset: Word prediction requiring a broad discourse context. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 1525–1534, 2016.

Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. Are emergent abilities of large language models a mirage? In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023. Best Paper Award.

Rico Sennrich, Barry Haddow, and Alexandra Birch. Neural machine translation of rare words with subword units. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 1715–1725, 2016.

Mengzhou Xia, Mikel Artetxe, Jingfei Du, Danqi Chen, and Veselin Stoyanov. Training trajectories of language models across scales. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023.

A Token-weighted vs word-occurrence-matched aggregation

Earlier drafts of this work used a "multi-token-strict" aggregation that restricted to multi-token words but then computed **token-weighted** class means: each continuation token contributed once, each initial token contributed once, but a word with 4 continuation subwords contributed 4 continuation observations and 1 initial observation. Adversarial review flagged that this does not give equal weight per word; longer-fragmented words dominate the continuation class.

We therefore adopted the **per-word matched** aggregation reported in the main paper: each multi-token word contributes one (init-loss, mean-cont-loss) pair, weighted equally.

Comparison on Pythia 7 non-anomalous sizes at D=300B, mid deciles 4–7:

Aggregation	α_{init}	α_{cont}	$\Delta\alpha$	95% CI
Token-weighted multi-token-strict	0.72	1.03	−0.30	$[-0.43, -0.16]$
Per-word matched (paper)	**0.56**	**0.98**	**−0.41**	$[\mathbf{-0.64}, \mathbf{-0.22}]$

Per-word matched aggregation gives larger $|\Delta\alpha|$ and a wider CI. The sign and qualitative conclusion are robust to the aggregation choice; the magnitude is sensitive. We report the per-word matched estimand throughout the main paper.

B Joint L(N, D) functional form survey

We attempted joint per-class $L_c(N, D)$ fits with 5 functional forms:

Form	Functional shape	# free params per class
ADD (Chinchilla-additive)	$E + A \cdot N^{-\alpha_N} + B \cdot D^{-\alpha_D}$	5
MULT (multiplicative)	$E + C \cdot N^{-\alpha_N} \cdot D^{-\alpha_D}$	4
HOFF (Hoffmann-generalized)	$E + (A \cdot N^{-\alpha_N} + B \cdot D^{-\alpha_D})^{\beta}$	6
COMPUTE (compute-only)	$E + K \cdot (N \cdot D)^{-\alpha}$	3
MIN (saturation)	$E + \min(A \cdot N^{-\alpha_N}, B \cdot D^{-\alpha_D})$	5

Estimation: grid search over the α exponents in $[0.02, 2.5]$, remaining parameters by Nelder–Mead with multi-start (30 random inits), enforcing $E \geq 0$, $A \geq 0$, $B \geq 0$.

Result: all five forms produce non-identifiable fits on our (N, D) range. The ADD form rails one of (α_N, α_D) at the grid edge; MULT collapses α_N to ~ 0; HOFF over-fits the 6 free parameters; MIN selects the larger of the two terms uniformly. The underlying issue is that Pythia's N range covers $\sim 12\times$ (1B $\rightarrow$ 12B excluding small N), while the D range covers $\sim 150\times$ (2B $\to$ 300B). The D-direction variation dominates the loss surface, so the optimizer absorbs per-class variation into the D-axis terms and the N-axis terms become weakly identified.

This is itself a methodological observation: **joint $L(N, D)$ fits require a balanced (N, D) range**. The Chinchilla scaling literature typically fits joint $L(N, D)$ on families with N range comparable to D range; Pythia's checkpoint-rich but N-narrow setup is the wrong substrate for joint identification.

We therefore report **per-axis fits** (one axis varied, the other fixed) as the primary methodology. The per-axis fits at D=300B (varying N) and at fixed $N \in \{1B, 1.4B, 6.9B, 12B\}$ (varying D) are the consistent decompositions we can support with our data.

C HuggingFace Hub model-file integrity bugs in Pythia checkpoints

During this work we identified two classes of HuggingFace Hub metadata corruption affecting Pythia intermediate-step checkpoints. All SHA256 hashes and sizes in this appendix are the values reported by the Hub itself in the `x-linked-etag` and `x-linked-size` HTTP response headers of the `/resolve/<revision>/<filename>` endpoint, which are server-side metadata — they reflect the LFS blob the Hub maps each revision to, independent of the `transformers` or `huggingface_hub` client resolution layer. The bug locus is therefore **Hub-side**, not client-side. The full audit (3 files $\times$ 8 revisions) is reproducible by `python3 verify_checkpoints.py` in the released repository, with the JSON output committed as `hub_blob_audit.json`.

C.1 Pythia-2.8B `model.safetensors` mis-registration

For the `EleutherAI/pythia-2.8b` repository, the `model.safetensors` file at multiple intermediate-step revisions has LFS blob SHA256 hashes identical to `main`'s, despite distinct Git commit SHAs at each revision (proving the Hub itself, not the client, is mapping these revisions to the main-weights blob):

revision	commit SHA[:12]	safetensors LFS SHA256[:16]	`pytorch_model.bin` LFS SHA256[:16]
`main`	`2a259cdd96a4`	`ab496f1c3fd79e3c`	`36f0f93d1f5d44ca`
`step1000`	`cc81f94ea2ac`	`ab496f1c3fd79e3c` (= main)	`36f0f93d1f5d44ca` (= main)
`step2000`	`7378edcedb6f`	`ab496f1c3fd79e3c` (= main)	`36f0f93d1f5d44ca` (= main)
`step4000`	`494474bbd952`	`ab496f1c3fd79e3c` (= main)	`36f0f93d1f5d44ca` (= main)
`step10000`	`835c7474e203`	`ab496f1c3fd79e3c` (= main)	`36f0f93d1f5d44ca` (= main)
`step50000`	`1936c7c29530`	`ab496f1c3fd79e3c` (= main, still wrong)	`73468bb90e734f88`
`step100000`	`668bcb3f3a13`	`ab496f1c3fd79e3c` (= main, still wrong)	`ced5e41d87bf8470`
`step143000`	`dbe7ae300a54`	`462f2b960062159c`	`a294bfcf6ff2c091`

Loading via the standard `AutoModelForCausalLM.from_pretrained(..., revision="step1000")` API silently returns the `main`-weights model because `transformers` prefers safetensors, and the safetensors file at step1000 is byte-identical to main's. The `pytorch_model.bin` files have correct per-revision hashes for step50000 and step100000 (but not for earlier steps). `use_safetensors=False` works for revisions where the `.bin` is correctly registered.

For early steps (1000–10000), both file formats point to `main`; the weights are unrecoverable from HF Hub via the standard API for those revisions. We exclude all Pythia-2.8B intermediate-step revisions from our D-axis fit and use only the final checkpoint for the N-axis fit and cross-family analysis.

C.2 Pythia-12B step50000 shard truncation

For the `EleutherAI/pythia-12b` repository, the `pytorch_model-00003-of-00003.bin` file at revision step50000 has Hub-reported size 3,896,183,087 bytes ($\sim$3.9 GB) compared to 4,105,939,923 bytes ($\sim$4.1 GB) at neighbouring revisions ($\sim$5.1% under-size, $\sim$200 MB missing):

revision	shard-3 LFS SHA[:16]	shard-3 size (bytes)	note
`main`	`eb82155af2c813a9`	4,105,939,923	
`step42000`	`541de6140dc4213c`	4,105,939,923	
`step50000`	`7df63b4231856f55`	3,896,183,087	truncated
`step70000`	`e731f1d1b670e975`	4,105,939,923	
`step100000`	`b625404aa7e3e793`	4,105,939,923	

Loading the step50000 model and running forward passes produces NaN losses (the truncated final shard appears to contain partial / malformed tensors). `use_safetensors=False` does not fix this. We exclude step50000 from the 12B D-axis fit; the 14 other dense checkpoints are unaffected.

C.3 Verification procedure

For any scaling-law study using Pythia intermediate checkpoints, before trusting the results we recommend:

1. For each revision used, compare model-file SHA256 to the `main` revision's. If equal, the revision is silently mis-registered.
2. Compare model-file size across revisions; truncation is detectable from size alone.
3. Run a sanity inference on the loaded model and check overall loss against expected (vs the Pythia paper's reported values per checkpoint, or vs an adjacent revision).
4. If using safetensors, also load with `use_safetensors=False` and compare; mismatch indicates HF Hub metadata corruption.

Verification scripts and the affected-revision list are at the project repository.